

NEO Mathematics Supporting Adult Guidance

Year 7 Place Value - How Much Is a Digit Worth?

Connection before curriculum. Always. Make the mathematics more enterable; do not increase pressure or take over the learner's thinking.

Lesson purpose

This lesson develops positional place value across integers and decimals. The mathematical spine is the multiplicative relationship between neighbouring columns: one place left is ten times the value; one place right is one tenth of the value.

Stepping Stones

Identify digit value; reason about factors of ten between neighbouring columns; explain placeholder zeros; repair misconceptions; and move between standard and expanded representations.

Suggested 45-minute route

Phase	Approx.	Supporting move
Arrive / Connect	5 min	Open only the first Connection panel. Ask what gives a digit its value.
Connection	7 min	Use the budget context if helpful. Keep the place-value relationship central.
Movement	15 min	Invite prediction before reveal. Ask what changed and what stayed the same.
Reflection	8 min	Treat mistakes as information. Repair one statement and explain the changed idea.
Creativity / Evidence	7 min	Build one strong museum exhibit rather than rushing through every requirement.
Rest / close	3 min	Use the one-column-at-a-time routine and name one next step.

Cornerstone guidance

Cornerstone	What to notice / support
Connection	Digit-position-value; neighbouring columns; accuracy and decisions.
Movement	The learner changes a digit or column and observes a mathematical response. Ask: What changed? What stayed the same?
Reflection	Use misconception repair and representation choice. Prefer: What idea might this answer be using?
Creativity	Museum work is teach-back. Poster, sketch, audio or digital product are legitimate.
Rest	One digit, one column, one neighbouring relationship.
Nutrition	Not forced in this lesson. No food or health context is required.

Likely misconceptions and useful prompts

Misconception	Prompt rather than tell
---------------	-------------------------

0.4 is smaller than 0.35 because $4 < 35$	Can we place both numbers on the same place-value chart? What is 0.4 in hundredths?
Moving one place changes value additively	Did we add a fixed amount, or did the value scale? Compare the two values.
Placeholder zero has no purpose	What happens to the other digits if the zero disappears?
The digit itself determines its value	Where is the digit? What would the same digit be worth one column left?

Using the NEO Maths Scratchpad

The Scratchpad is optional and stays inside the lesson. Encourage it for predictions, quick place-value columns, misconception repair and museum planning. Do not insist on neatness. Sticky Notes can hold a single rule or question. The working page can be stored in the learner's browser or saved as an image.

Supporting language

Helpful prompts: What do you notice? - Which column is the digit in? - What is the value, not just the digit? - What changed when it moved? - Can you show that another way? - Would a pause or one-column view help?

Resources and materials

The HTML lesson can run alone. Optional resources: paper and pencil; mini whiteboard; place-value chart; digit cards 0-9; decimal-point card; calculator for checking only; headphones for spoken explanation; saved Scratchpad image for portfolio evidence.

Evidence and mastery

Use the Final Task and lesson interactions as evidence across Understanding, Application, Communication and Reflection using Emerging, Developing, Secure and Mastering. A learner does not need to complete every activity to show strong evidence.

When to stop or adapt

Pause or reduce the visible task if the learner is scanning without engaging, repeatedly guessing, becoming verbally self-critical, or losing the position of the digit. Return to one digit and one neighbouring column. The mathematics can remain ambitious while the amount held in attention becomes smaller.